

Introduction to Trigonometric Identities

What is an Identity

DEFINITION: *Identity*

In mathematics, an identity is any equation that holds true for all values of the variable, provided those values are from the domain.

Examples: $\frac{2x}{4} = \frac{x}{2}$, $2(x + 4) = -6x + 4(2 + 2x)$, $\log(2x) = \log(2) + \log(x)$

On occasion, an identity will be something apparent, but they oftentimes (the more useful ones at least) express relationships that are not obvious. It can even take a great deal of effort to verify an equation is an identity, which we'll address later in this text. For now, we focus on the application of some of the more fundamental identities.

Ratio and Reciprocal Identities

We've informally encountered these identities before. We now provide a formal introduction.

PROPERTY: *Ratio and Reciprocal Identities*

Let θ be any value for which the following functions are defined.

Reciprocal Identities	$\sin(\theta) = \frac{1}{\csc(\theta)}$	$\csc(\theta) = \frac{1}{\sin(\theta)}$
	$\cos(\theta) = \frac{1}{\sec(\theta)}$	$\sec(\theta) = \frac{1}{\cos(\theta)}$
	$\tan(\theta) = \frac{1}{\cot(\theta)}$	$\cot(\theta) = \frac{1}{\tan(\theta)}$
Ratio Identities	$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$	$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$

Tips for recalling the ratio and reciprocal identities:

- ✓ Most students immediately recognize a relationship between tangent and cotangent. They're "cofunctions," so their names are quite similar, and grouping them together is a natural choice. But, be careful; we also have **cofunction identities** which are entirely separate from reciprocal identities here!
- ✓ Reciprocal functions are always paired with another function with **differing first letters**. For example, cosecant cannot be paired with cosine (both begin with the same letter), so cosecant and sine are paired together. Same for cosine and secant.

EXAMPLE 1

Given $\sin(\theta) = \frac{2\sqrt{6}}{7}$ and $\tan(\theta) = \frac{2\sqrt{6}}{5}$, find the remaining four trig ratios.

$$\sin \theta = \frac{2\sqrt{6}}{7} \rightarrow \csc \theta = \frac{7}{2\sqrt{6}} \left(\frac{\sqrt{6}}{\sqrt{6}} \right) = \frac{7\sqrt{6}}{12}$$

$$\tan \theta = \frac{2\sqrt{6}}{5} \rightarrow \cot \theta = \frac{5}{2\sqrt{6}} \left(\frac{\sqrt{6}}{\sqrt{6}} \right) = \frac{5\sqrt{6}}{12}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \rightarrow \cos \theta = \frac{\sin \theta}{\tan \theta} = \frac{2\sqrt{6}/7}{2\sqrt{6}/5} = \frac{2\sqrt{6}}{7} \cdot \frac{5}{2\sqrt{6}} = \frac{5}{7}, \cos \theta = \frac{5}{7}$$

$\hookrightarrow \sec \theta = \frac{7}{5}$

EXAMPLE 2

If $0^\circ \leq \theta < 360^\circ$, then there are exactly two θ -values for which $\sin(\theta) = 0$ and two for which $\cos(\theta) = 0$. In particular, $\sin(0^\circ) = \sin(180^\circ) = \cos(90^\circ) = \cos(270^\circ) = 0$. Given this information, your knowledge of ratio and reciprocal identities, and that $0^\circ \leq \theta < 360^\circ$, answer the following.

(a) For what values of θ is $\tan(\theta)$ not defined?

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \leftarrow \text{cannot} = 0!$$

$\tan \theta$ undefined for
 $\theta = 90^\circ$ and $\theta = 270^\circ$

(b) For what values of θ is $\cot(\theta)$ not defined?

$$\cot \theta = \frac{\cos \theta}{\sin \theta} \leftarrow \text{cannot} = 0!$$

$\cot \theta$ undefined for
 $\theta = 0^\circ$ and $\theta = 180^\circ$

(c) For what values of θ is $\sec(\theta)$ not defined?

$$\sec \theta = \frac{1}{\cos \theta} \leftarrow \text{cannot} = 0!$$

$\sec \theta$ undefined for
 $\theta = 90^\circ$ and $\theta = 270^\circ$

(d) For what values of θ is $\csc(\theta)$ not defined?

$$\csc \theta = \frac{1}{\sin \theta} \leftarrow \text{cannot} = 0!$$

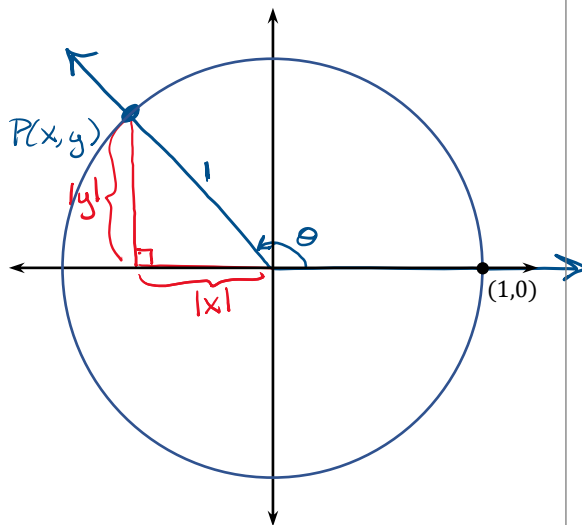
$\csc \theta$ undefined for
 $\theta = 0^\circ$ and $\theta = 180^\circ$

Pythagorean Identities

As you've probably guessed, the Pythagorean identities are related to the well-known Pythagorean theorem.

EXAMPLE 3

- Using the provided graph, pick any point along the circle and label this point as $P(x, y)$. (While it is acceptable to choose any point, for the purpose of this example, use a point not along the x - or y -axis.)
- Construct an angle θ , in standard position, passing through point P .
- Find $\sin(\theta)$ and $\cos(\theta)$.



$$\sin \theta = \frac{y}{r} = \frac{y}{1} \rightarrow y = \sin \theta$$

$$\cos \theta = \frac{x}{r} = \frac{x}{1} \rightarrow x = \cos \theta$$

- What do we know about the quantity, $x^2 + y^2$? (**Hint:** Draw a right triangle using the terminal side of theta as the hypotenuse and base along the x -axis.)

$$(|x|)^2 + (|y|)^2 = 1^2 \text{ (Pyth. thm)} \rightarrow x^2 + y^2 = 1$$

- Insert your findings from (c) into (d); you have just demonstrated the Pythagorean identity!

$$x^2 + y^2 = 1 \rightarrow (\cos \theta)^2 + (\sin \theta)^2 = 1 \rightarrow \cos^2 \theta + \sin^2 \theta = 1$$

- This circle, referred to as the **unit circle**, provides an excellent visual for understanding the domain and range for both the sine and cosine functions. Considering this graph, determine an appropriate domain and range for the sine and cosine functions.

θ is an angle representing any amount of rotation! $\theta \in (-\infty, \infty)$ domain ↗

$y = \sin \theta$ with $-1 \leq y \leq 1 \rightarrow \sin \theta \in [-1, 1]$ range for sine

$x = \cos \theta$ with $-1 \leq x \leq 1 \rightarrow \cos \theta \in [-1, 1]$ range for cosine

PROPERTY: Pythagorean Identities

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$\sin^2(\theta) + \cos^2(\theta) = 1 \quad \leftarrow \text{only need to remember this...}$$

$$\tan^2(\theta) + 1 = \sec^2(\theta)$$

$$\cot^2(\theta) + 1 = \csc^2(\theta) \quad \frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$

EXAMPLE 4

Let θ be an angle in quadrant III such that $\cos(\theta) = -\frac{3}{4}$. Find $\sin(\theta)$ and $\cot(\theta)$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\sin \theta = \pm \sqrt{1 - \cos^2 \theta}$$

$$\sin \theta = -\sqrt{1 - \cos^2 \theta}$$

$\uparrow$ $\theta \in \text{Q III}$

$$\sin \theta = -\sqrt{1 - (-3/4)^2}$$

$$= -\sqrt{1 - 9/16}$$

$$= -\sqrt{7/16}$$

$$\sin \theta = -\frac{\sqrt{7}}{4}$$

$$\cot \theta = \frac{-3/4}{-\sqrt{7}/4} = \frac{3}{\sqrt{7}} \left(\frac{\sqrt{7}}{\sqrt{7}} \right) = \frac{3\sqrt{7}}{7}$$

EXAMPLE 5

On her trigonometry exam, Ariel was given that $\tan(\theta) = \frac{3}{2}$ and asked to find $\sin(\theta)$ and $\cos(\theta)$. She claimed that $\sin(\theta) = 3$ and $\cos(\theta) = 2$. She was awarded a small amount of partial credit for this response but not full marks. What does this answer reveal about her understanding, and why was she not awarded full credit? Determine the correct response.

$$\tan \theta = \frac{3}{2} = \frac{\sin \theta}{\cos \theta} \quad \text{does not imply } \sin \theta = 3, \cos \theta = 2$$

$\uparrow$ not possible!

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\sec \theta = \pm \sqrt{\tan^2 \theta + 1}$$

$$= \pm \sqrt{(3/2)^2 + 1}$$

$$= \pm \sqrt{9/4 + 1}$$

$$= \pm \sqrt{13/4}$$

$$= \pm \frac{\sqrt{13}}{2}$$

$$\sec \theta = \pm \frac{\sqrt{13}}{2} \rightarrow \cos \theta = \pm \frac{2}{\sqrt{13}} = \pm \frac{2\sqrt{13}}{13}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \rightarrow \sin \theta = \cos \theta \tan \theta$$

$$\sin \theta = \pm \frac{2\sqrt{13}}{13} \left(\frac{3}{2} \right)$$

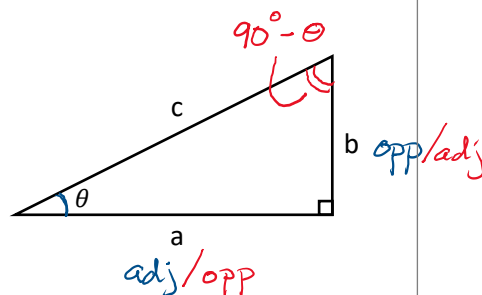
$$\sin \theta = \pm \frac{3\sqrt{13}}{13}$$

EXAMPLE 6

The Cofunction Identities: You've probably wondered about the relationship between sine & cosine, secant & cosecant, as well as tangent & cotangent. Let's develop an understanding of these cofunctions by way of example.

(a) Consider the triangle provided here. Label the missing angle in terms of θ .

(b) Evaluate $\sin(\theta)$, then restate this value in terms of cosine alone. [Hint: Consider using the missing angle you labeled in (a).]



$$\sin \theta = \frac{b}{c} = \frac{b}{c} = \cos(90^\circ - \theta)$$

(c) Evaluate $\sec(\theta)$, then restate this value in terms of cosecant alone.

(d) Evaluate $\tan(\theta)$, then restate this value in terms of cotangent alone.

$$\sec \theta = \frac{c}{a} = \frac{c}{a} = \csc(90^\circ - \theta) \quad \tan \theta = \frac{b}{a} = \frac{b}{a} = \cot(90^\circ - \theta)$$

(e) For any acute angle θ , summarize your findings from above in the provided table.

$\sin(\theta) = \cos(90^\circ - \theta)$	$\cos(\theta) = \sin(90^\circ - \theta)$
$\sec(\theta) = \csc(90^\circ - \theta)$	$\csc(\theta) = \sec(90^\circ - \theta)$
$\tan(\theta) = \cot(90^\circ - \theta)$	$\cot(\theta) = \tan(90^\circ - \theta)$

Note: These angles are complementary!
 $\theta + (90^\circ - \theta) = 90^\circ$

(f) Write $\sec(10^\circ)$ in terms of its cofunction.

$$\sec(10^\circ) = \csc(90^\circ - 10^\circ)$$

$$\sec(10^\circ) = \csc(80^\circ)$$

(g) Solve for x .

$$\sin(\underbrace{x + 8^\circ}_{\theta}) = \cos(\underbrace{40^\circ + 2x}_{90^\circ - \theta})$$

Complementary!

$$(x + 8) + (40 + 2x) = 90$$

$$3x + 48 = 90$$

$$3x = 42$$

$$x = 14$$

Check $\sin(14 + 8) = \cos(40 + 2(14))$

$$\sin(22) = \cos(68)$$

✓

